

Tanaka Certificates

July 1, 2026

Abstract

In these notes, I am building up the intuition and theory behind supermartingale certificates, with relaxed assumptions on the certificate function V . In particular, it is explored what happens when V is not twice continuously-differentiable, but instead piecewise twice continuously-differentiable.

1 Piecewise linear certificates in dimension one

Consider the following setup:

Setup

Let:

1. $dX_t = f(X_t)dt + g(X_t)dW_t$ be a one-dimensional SDE.
2. $V(x)$ be a piecewise linear function such that $V(x) = a_i x + b_i$, where $x \in (x_i, x_{i+1})$.
3. $V(x)$ is continuous at breakpoints: $V(x_i^+) = V(x_i^-) = V(x_i)$
4. as a consequence, $V'_-(x_i) = a_{i-1}$, $V'_+(x_i) = a_i$

Now we will consider what conditions on V are sufficient to guarantee that $V(X_t)$ is a supermartingale.

Hypothesis

Does concavity of $V(x)$ (along with an interior / drift term condition) guarantee that $V(X_t)$ is a supermartingale?

To answer this, let's apply the Itô-Tanaka formula to $V(X_t)$:

$$V(X_t) = V(X_0) + \int_0^t V'(X_s) f(X_s) ds \quad (1)$$

$$+ \int_0^t V'(X_s) g(X_s) dW_s \quad (2)$$

$$+ \frac{1}{2} \sum_{i=1}^{n-1} (V'_+(x_i) - V'_-(x_i)) L_t^{x_i}(X) \quad (3)$$

$$(4)$$

We will focus on these terms:

1. $D_t = \int_0^t V'(X_s) f(X_s) ds$, the drift term
2. $M_t = \int_0^t V'(X_s) g(X_s) dW_s$, the diffusion term (stochastic integral)
3. $K_t = \frac{1}{2} \sum_{i=1}^{n-1} (V'_+(x_i) - V'_-(x_i)) L_t^{x_i}(X)$, the kink term

we need to prove that $D_t + M_t + K_t$ is a supermartingale.

Some natural assumptions, which are sufficient to guarantee that $V(X_t)$ is a supermartingale, are the following:

1. $f(x)V'(x) \leq 0$ for all $x \in (x_i, x_{i+1})$. This is because of the following:

Fact

If $f(X)V'(X) \leq 0$, then the drift term D_t is a supermartingale.

Proof: We want to prove $\mathbb{E}[D_t|\mathcal{F}_s] \leq D_s$ for $s \leq t$.

$$\mathbb{E}[D_t|\mathcal{F}_s] = \mathbb{E} \left[\int_0^t V'(X_u)f(X_u)du \middle| \mathcal{F}_s \right] \quad (5)$$

$$= \int_0^s V'(X_u)f(X_u)du + \mathbb{E} \left[\int_s^t V'(X_u)f(X_u)du \middle| \mathcal{F}_s \right] \quad (6)$$

$$\leq D_s + 0 = D_s \quad (7)$$

2. M_t is a martingale. Assuming $V(x)$ has finitely many pieces (as is the case for a neural network with ReLU activation functions), it is enough to impose a square-integrability condition on $g(x)$ ¹
3. $V'_+(x_i) - V'_-(x_i) \leq 0 \iff a_{i-1} \geq a_i$, i.e. the **concavity condition**. This is because of the following:

Fact

If $V'_+(x_i) - V'_-(x_i) \leq 0$ and $L_t^{x_i}(X) \geq 0$, then the kink term K_t is a supermartingale.

Proof: We want to prove $\mathbb{E}[K_t|\mathcal{F}_s] \leq K_s$ for $s \leq t$.

$$\mathbb{E}[K_t|\mathcal{F}_s] = \mathbb{E} \left[\frac{1}{2} \sum_{i=1}^{n-1} (V'_+(x_i) - V'_-(x_i)) L_t^{x_i}(X) \middle| \mathcal{F}_s \right] \quad (8)$$

$$= \frac{1}{2} \sum_{i=1}^{n-1} (V'_+(x_i) - V'_-(x_i)) \mathbb{E}[L_t^{x_i}(X)|\mathcal{F}_s] \quad (9)$$

$$\leq \frac{1}{2} \sum_{i=1}^{n-1} (V'_+(x_i) - V'_-(x_i)) L_s^{x_i}(X) = K_s \quad (10)$$

Note that the inequality follows from the fact that $L_t^{x_i}(X)$ is non-decreasing in t , hence $\mathbb{E}[L_t^{x_i}(X)|\mathcal{F}_s] \geq \mathbb{E}[L_s^{x_i}(X)|\mathcal{F}_s] = L_s^{x_i}(X)$.

We will consider what happens in different practical cases to these terms, including:

1. A simple one-dimensional SDE with a piecewise linear certificate.
2. A pendulum with a stochastic torque, and X_t measuring the angle of the pendulum.
3. A network b black and w white nodes, with given transition probabilities, and X_t measuring the amount of mass in the white nodes.

to ensure our assumptions are reasonable.

1.1 Proof of the supermartingale property

First, let's note the following:

¹For example, $\mathbb{E} \left[\int_0^t g(X_s)^2 ds \right] < \infty$.

Fact

The sum of a martingale and a supermartingale is a supermartingale.

Proof: By definition, given a martingale M_t and a supermartingale S_t , for $r < t$ we have $\mathbb{E}[M_t|\mathcal{F}_r] = M_r$ and $\mathbb{E}[M_t|\mathcal{F}_r] \leq S_r$. Then, for $r < t$:

$$\mathbb{E}[M_t + S_t|\mathcal{F}_r] = \mathbb{E}[M_t|\mathcal{F}_r] + \mathbb{E}[S_t|\mathcal{F}_r] \quad (11)$$

$$= M_r + \mathbb{E}[S_t|\mathcal{F}_r] \quad (12)$$

$$\leq M_r + S_r = (M + S)_r \quad (13)$$

hence $M_t + S_t$ is a supermartingale.

The drift term D_t is a supermartingale by assumption 1. The kink term K_t is a supermartingale by assumption 3. The stochastic integral M_t is a martingale by assumption 2. Therefore, the sum of these three terms is a supermartingale (as a consequence of the previous fact), and hence $V(X_t)$ is a supermartingale.

2 Piecewise $\mathcal{C}^2$ certificates in dimension one

The setup is pretty similiar, with the exception that V is now piecewise $\mathcal{C}^2$ instead of piecewise linear.

Setup

Let:

1. $dX_t = f(X_t)dt + g(X_t)dW_t$ be a one-dimensional SDE.
2. $V(x)$ be a piecewise $\mathcal{C}^2$ function such that $V(x) \in \mathcal{C}^2$, where $x \in (x_i, x_{i+1})$.
3. $V(x)$ is continuous at breakpoints: $V(x_i^+) = V(x_i^-) = V(x_i)$.

We will again consider what conditions on V are sufficient to guarantee that $V(X_t)$ is a supermartingale. Again, the natural hypothesis is:

Hypothesis

Does concavity of $V(x)$ (along with an interior / drift term condition) guarantee that $V(X_t)$ is a supermartingale?

We apply the Itô-Tanaka formula to $V(X_t)$, and we get similiar terms, with the addition of

$$C_t = \frac{1}{2} \int_0^t V''(X_s) g^2(X_s) ds \quad (14)$$

Since the integral in the C_t term is over the same intervals as the D_t term, we can combine them to recover the generator / Lie derivative term:

$$D_t + C_t = \int_0^t \left(V'(X_s) f(X_s) + \frac{1}{2} V''(X_s) g^2(X_s) \right) ds = \int_0^t \mathcal{L}V(X_s) ds \quad (15)$$

The proof that $V(X_t)$ is a supermartingale is similiar to the piecewise linear case, but now we need to impose the condition:

$$\mathcal{L}V(x) = V'(x)f(x) + \frac{1}{2} V''(x)g^2(x) \leq 0 \quad (16)$$

Table 1: Piecewise linear certificates examples.

SDE	$V(x)$	D_t	M_t	K_t
$dX_t = dW_t$	$- x $	0	$\int_0^t -\text{sgn}(X_s) dW_s$	$-L_t^0(X)$
$dX_t = -X_t dt$	$ x $	$-\int_0^t X_s ds$	0	0

Table 2: Piecewise $\mathcal{C}^2$ certificates examples.

SDE	$V(x)$	D_t	M_t	C_t	K_t
$dX_t = -\lambda X_t dt + \sigma dW_t$	x^2	$-2\lambda \int_0^t X_s^2 ds$	$\int_0^t 2\sigma X_s dW_s$	$\sigma^2 t$	0
$dX_t = dt + \sqrt{2} dW_t$	$x - \frac{x^2}{2}$	$\int_0^t (1 - X_s) ds$	$\int_0^t (1 - X_s) \sqrt{2} dW_s$	$-t$	0
$dX_t = dW_t$	$\begin{cases} -\frac{1}{2}x^2, & x < 0, \\ -x, & x \geq 0 \end{cases}$	0	$\int_0^t V'(X_s) dW_s$	$-\frac{1}{2} \int_0^t \mathbf{1}_{\{X_s < 0\}} ds$	$-\frac{1}{2} L_t^0(X)$

3 Studying piecewise certificates on concrete one-dimensional examples

Some useful examples to study are given in Tables 1 and 2.

3.1 Ornstein–Uhlenbeck process in three parameter regimes

Consider the Ornstein–Uhlenbeck process

$$dX_t = -\lambda X_t dt + \sigma dW_t$$

with the quadratic certificate $V(x) = x^2$. Its generator is

$$\mathcal{L}V(x) = -2\lambda x^2 + \sigma^2 = -2\lambda \left(x^2 - \frac{\sigma^2}{2\lambda} \right).$$

Consequently, if

$$r = \frac{\sigma}{\sqrt{2\lambda}},$$

then the generator is negative for $|x| > r$, zero for $|x| = r$, and positive for $|x| < r$. The decomposition is

$$X_t^2 - X_0^2 = \underbrace{-2\lambda \int_0^t X_s^2 ds}_{D_t} + \underbrace{2\sigma \int_0^t X_s dW_s}_{M_t} + \underbrace{\sigma^2 t}_{C_t},$$

with $K_t = 0$.

Figure 1 fixes $\lambda = 1$ and $X_0 = 1$, and uses $r = 0.5$, $r = 1$, and $r = 1.5$. These choices make the initial generator negative, zero, and positive, respectively. Moreover,

$$\mathbb{E}[X_t^2] = r^2 + (X_0^2 - r^2)e^{-2\lambda t},$$

so the expected generator retains the corresponding sign. In the middle case, X_t^2 is constant in expectation, but it is not a martingale: the pointwise generator vanishes only when $|X_t| = r$.

The stochastic integral M_t need not visibly oscillate on every individual sample path; the martingale statement is instead that its conditional drift is zero. Figure 1 therefore plots Monte Carlo expectations. The mean of M_t remains statistically consistent with zero, while D_t and C_t determine whether $\mathbb{E}[X_t^2 - X_0^2]$ decreases, remains zero, or increases.

3.2 Failure of piecewise linearity and success of curvature

For $f = 1$ and $g = \sqrt{2}$ on $[0, 1]$, consider a case where want to find a certificate V such that $V(0) = 0$ and $V(1) = 1$.

OU process with $V(x) = x^2$, $\lambda = 1$, and $X_0 = 1$

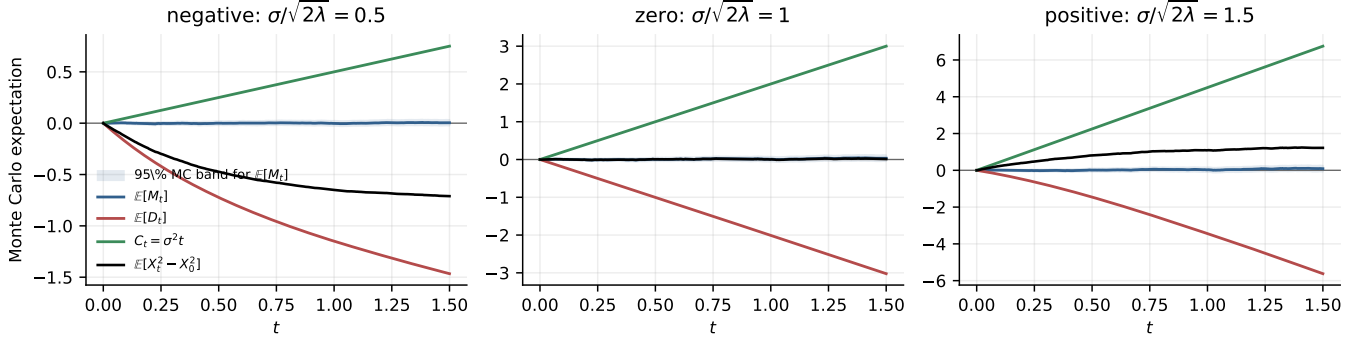


Figure 1: Monte Carlo expectations from 8000 Ornstein-Uhlenbeck paths for $V(x) = x^2$. The three columns use $\sigma/\sqrt{2\lambda} = 0.5, 1, 1.5$. The shaded region is a pointwise approximate 95% Monte Carlo band for $\mathbb{E}[M_t]$.

A smooth candidate could be:

$$V_{\mathcal{C}^2}(x) = x - \frac{x^2}{2},$$

the curvature contribution compensates for the positive drift:

$$\mathcal{L}V_{\mathcal{C}^2}(x) = (1 - x) + \frac{1}{2}(-1)(2) = -x \leq 0.$$

and indeed $V_{\mathcal{C}^2}$ is a valid supermartingale certificate.

On the other hand, let's assume V is piecewise linear. Then there exists an interval $(x_i, x_{i+1}) \subset [0, 1]$ where $V'(x) = a_i > 0$. On that interval, given an $x \in (x_i, x_{i+1})$, the curvature term vanishes, and the generator is

$$\mathcal{L}V(x) = V'(x)f(x) + \frac{1}{2}V''(x)g^2(x) = V'(x)f(x) = V'(x) = a_i > 0.$$

This example motivates the following hypothesis, which we will study in more detail in [Section 4](#).

Hypothesis

While piecewise linear certificates V_{PL} are not sufficient to guarantee that V_{PL} is a supermartingale, **piecewise quadratic certificates** are sufficient, as the curvature term can compensate for positive drift.

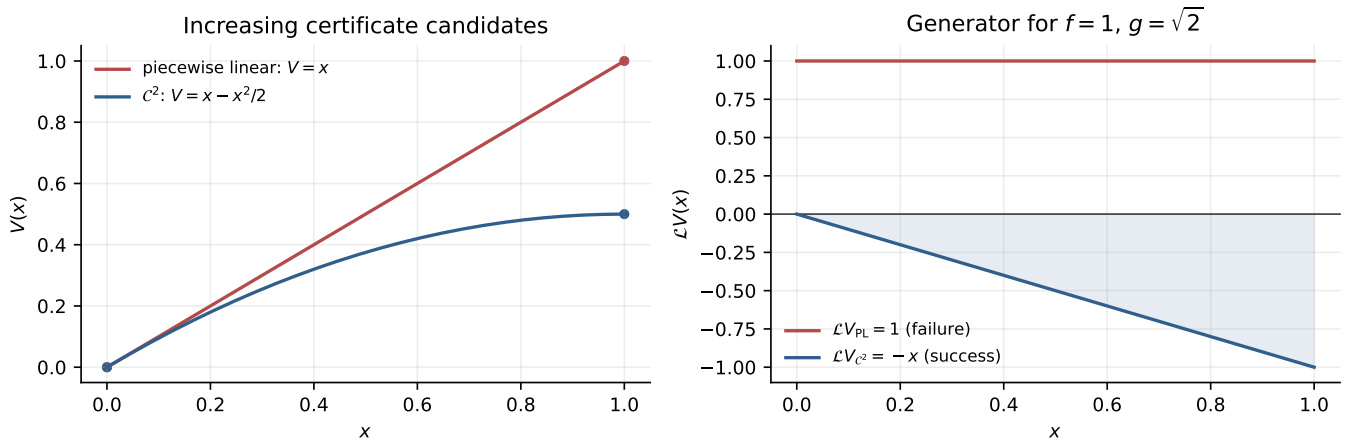


Figure 2: An increasing piecewise-linear candidate fails the generator condition for $f = 1$, $g = \sqrt{2}$, whereas negative curvature makes the smooth candidate satisfy it throughout $[0, 1]$.

3.3 A piecewise $\mathcal{C}^2$ certificate with curvature and a kink

Finally, let $dX_t = dW_t$ and

$$V(x) = \begin{cases} -\frac{1}{2}x^2, & x < 0, \\ -x, & x \geq 0. \end{cases}$$

Here $D_t = 0$, while

$$M_t = \int_0^t V'(X_s) dW_s, \quad C_t = -\frac{1}{2} \int_0^t \mathbf{1}_{\{X_s < 0\}} ds, \quad K_t = -\frac{1}{2} L_t^0(X).$$

Thus C_t decreases while the path lies below zero, and K_t decreases only when local time accumulates at the kink. Figure 3 shows these different mechanisms on the same Brownian realization. The plotted local time uses an occupation-density approximation with bandwidth 0.06.

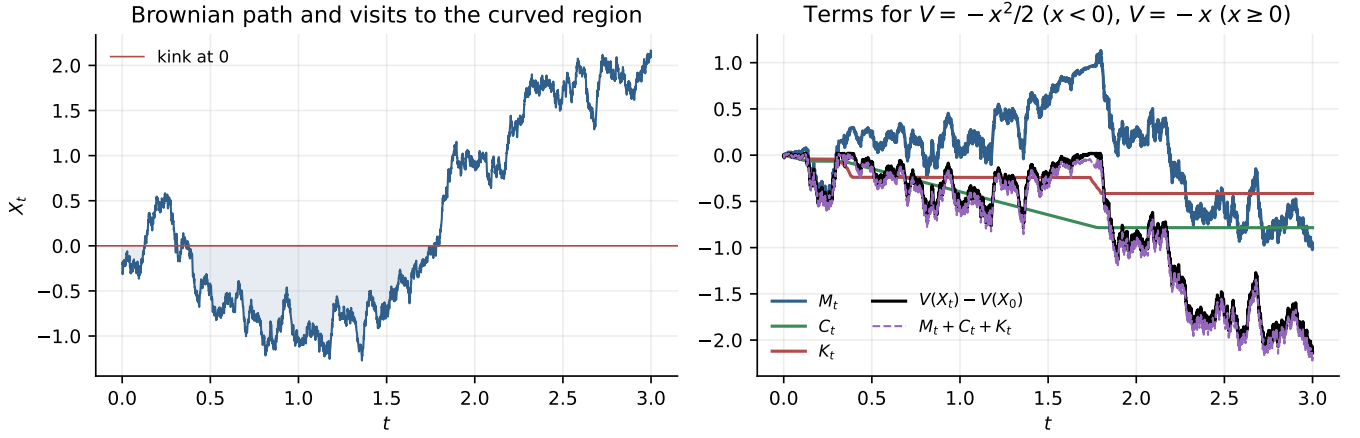


Figure 3: A Brownian realization and the terms for the hybrid piecewise $\mathcal{C}^2$ certificate. The curvature and kink terms are both nonincreasing, while the martingale term fluctuates.

4 Piecewise quadratic certificates in dimension one